Applying the Daubert Standard to Forensic Evidence (5e)

Digital Forensics, Investigation, and Response, Fifth Edition - Lab 01

Introduction

Established in 1993 as a result of the Supreme Court case *Daubert v. Merrell Dow Pharmaceuticals, Inc.* (509 U.S. 579), the *Daubert* standard is the test currently used in federal and most state courts to determine the admissibility of expert witness testimony and scientific evidence. Because the *Daubert* standard requires that scientific evidence presented in court be generally accepted in the field, new forensic tools may not be immediately approved for use in legal proceedings. However, emerging technologies, such as artificial intelligence (AI)-driven forensic analysis, blockchain for evidence integrity, and advanced malware detection, are rapidly shaping the future of digital forensics. Staying updated on these advancements is crucial for forensic investigators to ensure they are prepared to adopt validated tools as they gain acceptance, improve investigative accuracy, and remain competitive in the field.

The *Daubert* standard is crucial in digital forensics because it ensures that forensic tools and methodologies used in court are scientifically valid and reliable. In digital investigations, evidence from sources like hard drives, network logs, or mobile devices must be collected and analyzed using forensically sound techniques that meet legal standards. By applying the *Daubert* standard in labs, you will develop the skills needed to handle digital evidence professionally and ensure that forensic work stands up in court.

In this lab, you will act as a Digital Forensic Specialist assisting the lead Forensics Investigator at the Cyber Crimes Division (CCD) of the Boston Police Department. You have been given a hard drive image taken from a seized computer that is suspected to contain evidence of a sophisticated drug trafficking operation. First, you will review the search warrant and complete the Chain of Custody form that accompanies the evidence drive. You will then prepare the contents of the seized hard drive image as evidence in accordance with the *Daubert* standard using a variety of forensic tools.

Lab Overview

SECTION 1 of this lab has three parts, which should be completed in the order specified.

1. In the first part of the lab, you will review a search warrant and complete a Chain of Custody (CoC) form for seized evidence. This part of the lab will increase your understanding of search warrants and CoC procedures, which is essential for maintaining the integrity and admissibility of digital evidence in court.

2. In the second part of the lab, you will use FTK Imager to create hash codes for suspicious files. This part of the lab is critical because hashing is a fundamental part of digital forensics that ensures the integrity and authenticity of evidence.

3. In the third part of the lab, you will validate the hash codes using Paraben's Electronic Evidence Examiner (E3). This part of the lab will show you the importance of hash validation in digital forensics and reinforce

a key forensic principle: if the hash changes, the evidence is compromised.

SECTION 2 of this lab allows you to apply what you learned in **SECTION 1** with less guidance and different deliverables, as well as some expanded tasks and alternative methods. You will explore the suspect's drive image to locate additional evidence. You will also use Autopsy, another popular digital forensics tool, to validate the hash codes created during your preliminary investigation.

Finally, you will explore the virtual environment on your own in **SECTION 3** of this lab to answer a set of questions and challenges that allow you to use the skills you learned in the lab to conduct independent, unguided work—similar to what you will encounter in a real-world situation.

Learning Objectives

Upon completing this lab, you will be able to:

1. Prepare evidence for court and complete forms used in evidence handling.
2. Understand how a judge will determine the admissibility of evidence using the *Daubert* standard.
3. Import a drive image as evidence using digital forensics software.
4. Identify suspicious files as evidence using digital forensics software.
5. Create hash codes to verify the integrity of forensic evidence.

Topology

This lab contains the following virtual machines. Please refer to the network topology diagram below.

- vWorkstation (Windows: Server 2019)



Tools and Software

The following software and/or utilities are required to complete this lab. Students are encouraged to explore the Internet to learn more about the products and tools used in this lab.

- FTK Imager
- Paraben's E3
- Autopsy

Deliverables

Upon completion of this lab, you are required to provide the following deliverables to your instructor:

SECTION 1

1. Lab Report file, including screen captures of the following:
    - Contents of the search warrant in Adobe Reader
    - Completed Chain of Custody form in Adobe Reader
    - Contents of the 0002665_hash.csv file
    - Contents of the RecycleBinEvidence_hash.csv file
    - Contents of the MyRussianMafiaBuddies_hash.csv file
    - Contents of the Nice guys_hash.csv file
    - MD5 and SHA1 values for the MyRussianMafiaBuddies.txt file
    - MD5 and SHA1 values for the Nice Guys.png file
2. Any additional information as directed by the lab:
    - Describe how the hash values produced by E3 for the incriminating files compare to those produced by FTK. Do they match?

SECTION 2

1. Lab Report file, including screen captures of the following:

- Contents of the suspicious email file in the Display pane
- Two hash values for the suspicious email file
- MD5 field in the Result Viewer
- MD5 value produced by E3.

2. Any additional information as directed by the lab:

- Describe how the hash value produced by Autopsy compares to the values produced by FTK Imager for the two .eml files.
- Describe how the hash value produced by E3 compares to the values produced by FTK Imager for the two .eml files and the value produced by Autopsy.

SECTION 3

1. Lab Report file, including screen captures of the following:

- Hash values for the Evidence_drive1.001 file

2. Any additional information as directed by the lab:

- Define the original file names and file paths for each of the three files.

Hands-On Demonstration

Part 1: Complete Chain of Custody Procedures

The setting for this lab is an ongoing investigation into the activities of Beverly Gates, the HR Manager at Intricate Solutions, Inc. Senior leadership at Intricate Solutions has reason to believe that Beverly is involved in a sophisticated drug trafficking operation and has recently contacted the local police department to investigate the matter further. The police officer assigned to the case, Brendan O'Rourke, has already processed a search warrant and seized a hard drive at Beverly's home. The police department has assigned 10001-BPD-CCD as the case number for the Beverly Gates case. As a Digital Forensics Specialist assigned to the case with the Cyber Crimes Division (CCD) of the Boston Police Department, you will further investigate the seized hard drive.

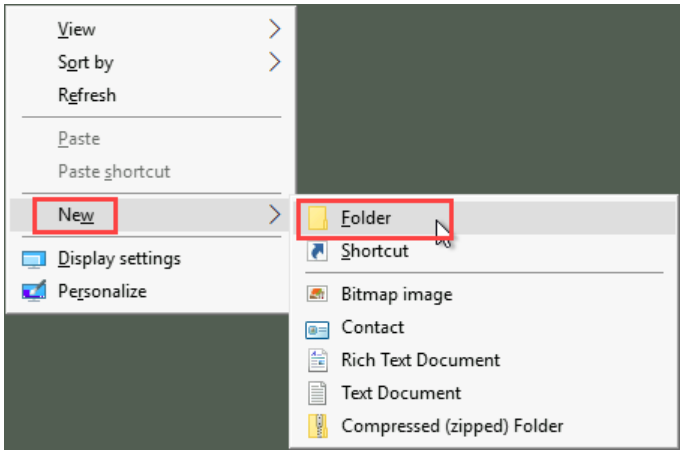
A search warrant is a legal document issued by a judge that authorizes law enforcement to search a specific location and seize designated evidence. When a search warrant is issued, it is critically important to observe the specific boundaries around what you are allowed to examine or seize. Overstepping those boundaries can jeopardize the entire investigation and render the evidence inadmissible in court.

In conjunction with the search warrant, the chain of custody process will determine whether evidence may be considered admissible in court. The chain of custody (CoC) refers to the chronological documentation that records the handling, transfer, and storage of evidence from the time it is collected until it is presented in court. If the CoC is not followed and the appropriate documentation not produced, the opposing attorney can challenge or dismiss the evidence presented. CoC is essential in ensuring that the seized hard drive remains authentic, untampered with, and legally admissible in court. Because digital evidence is easily altered, strict documentation and evidence-handling protocols must be followed to maintain its integrity.

Following these protocols ensures that the evidence you handle is both legally obtained and properly documented. This includes accurately completing forms that track every step in the evidence handling process, from seizure to transfer to final analysis. Properly handling and documenting the hard drive in this scenario protects the integrity of the investigation and helps ensure that any findings you present can be trusted in a court of law.

In the next steps, you will review the search warrant and complete the CoC form to take possession of the seized disk image.

1. On the vWorkstation desktop, **right-click** anywhere and **select New > Folder** from the context menu to create a new folder.



Create a new folder

2. In the folder name field, **type `10001-BPD-CCD`** and **press Enter** to name the new folder.

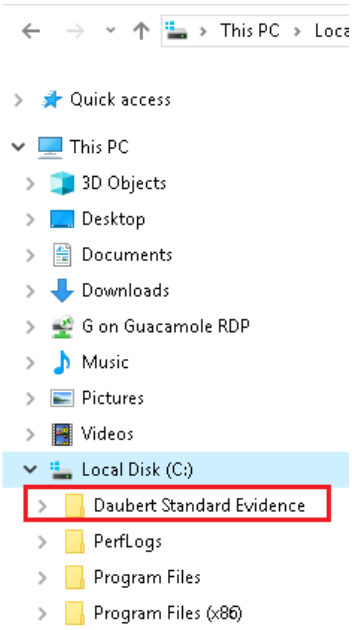
Note: You will save your work to this folder as the lab progresses.

3. On the vWorkstation taskbar, **click** the **File Explorer icon** to open a new File Explorer window.



File Explorer

4. In the File Explorer, **navigate** to **C:\Daubert Standard Evidence**.



Location of the target folder

5. In the Daubert Standard Evidence folder, **double-click** the **BG_warrant PDF** to open it in Adobe Reader.

6. In the Adobe Reader window, **review** the contents of the **BG_warrant PDF.**

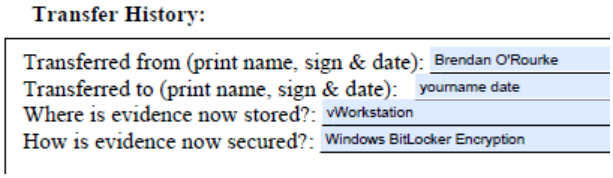
7. **Make a screen capture** showing the **contents of the search warrant in Adobe Reader**.

8. **Close** the **Adobe Reader window**.

9. In the Daubert Standard Evidence folder, **double-click** the **chain_of_custody_10001 PDF** to open it in Adobe Reader.

10. In the Transfer History section of the form, **type** the following information in the first block of entries to record the transfer of data.

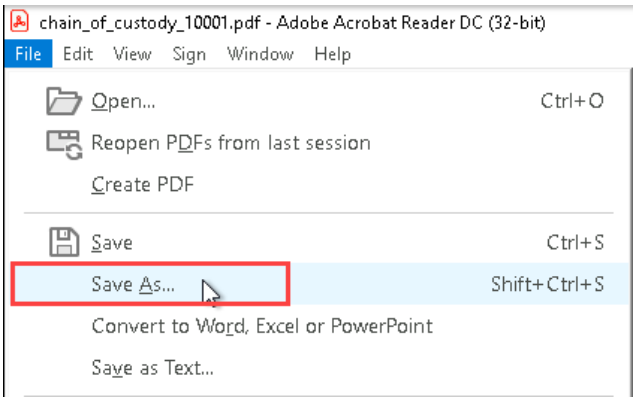
    - Transferred from: **`Brendan O'Rourke`**
    - Transferred to: ***`yourname`*** and ***`date`***, replacing yourname and date with your own name and the current date
    - Where is the evidence now stored: **`vWorkstation`**
    - How is the evidence now secured: **`Windows BitLocker Encryption`**



Transfer History

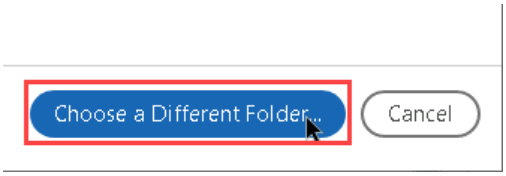
Note: Typically, this document would be signed by the investigator as well, but you will not be able to add an actual signature to this form.

11. In the Adobe Reader menu bar, **click File** and **select Save As** from the drop-down menu.



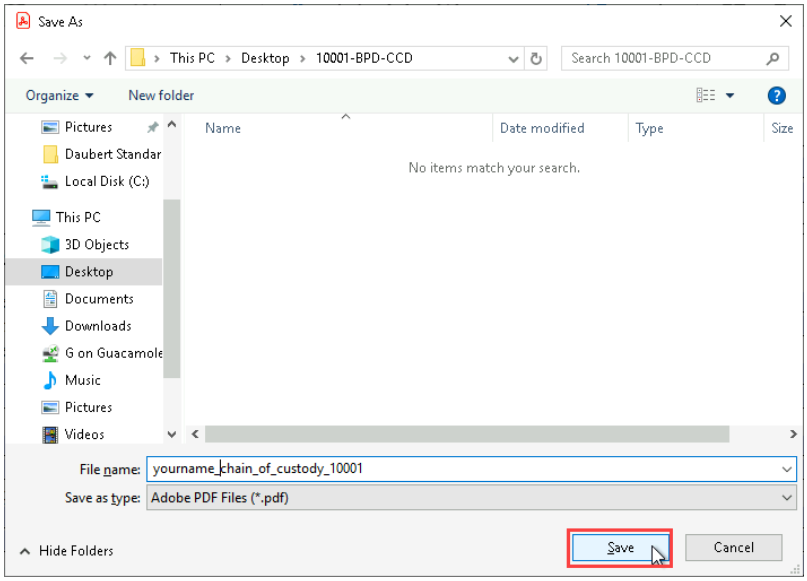
Save As menu entry

12. In the Save As dialog box, **click** the **Choose a Different Folder... button** to open another Save As dialog box.



Choose a Different Folder button

13. In the second Save As dialog box, **navigate** to the 10001-BBD-CCD folder you created earlier in the lab (**This PC > Desktop > 10001-BPD-CCD**), then **type** ***yourname*_chain_of_custody_10001** in the File name field (replacing *yourname* with your own name), and then **click Save** to save this file to the evidence folder.



Save As dialog box

14. **Make a screen capture** showing the **completed Chain of Custody form in Adobe Reader.**

15. **Close** the **Adobe Reader** and **File Explorer windows**.

Part 2: Extract Evidence Files and Create Hash Codes with FTK Imager

To adhere to the *Daubert* standard, it's crucial that all evidence in a digital investigation remains unchanged from the moment it is seized until it is presented in court. This is where hash codes come into play, acting as a digital fingerprint of a file. A hash code is a unique alphanumeric string generated by applying a cryptographic algorithm, such as MD5 or SHA1, to a file's contents. MD5 (Message-Digest Algorithm 5) produces a 128-bit hash value and is widely used for file integrity checks, although it is now considered less secure for cryptographic purposes due to vulnerabilities. SHA1 (Secure Hash Algorithm 1) produces a 160-bit hash value and is more secure than MD5, but it has also been found to have potential weaknesses against collision attacks. Even a small change to the file will produce a completely different hash code, making both algorithms useful for verifying data integrity. Just as fingerprints are unique to individuals, each file has its own unique hash code. This is a string of characters generated by a specific algorithm (e.g., MD5, SHA-1, or SHA-256) that represents the exact contents of the file.

When a file is first obtained, a hash code is created for it. If the file is later altered in any way, whether intentionally or accidentally, the hash code will change. By comparing the hash code of a file at different stages (e.g., when it's initially seized, when it's analyzed, and when it's presented in court), investigators can confirm whether the file has been tampered with.

In digital forensics, tools like FTK Imager are commonly used to generate these hash codes for every file extracted from a seized device. FTK Imager is a trusted, scientifically validated tool, ensuring that the process of evidence extraction follows the *Daubert* standard. By creating hash codes for each extracted file, you are verifying that the files you're working with are exact copies of the originals, without any changes.

As you proceed in this lab, you will not only extract files but also create and validate their hash codes. This ensures the integrity of the evidence, providing proof that no modifications have been made to the files during your analysis. This process is essential because the integrity of digital evidence can be easily challenged in court. By using validated forensic tools and following these procedures, you are ensuring the scientific reliability of the evidence, helping it withstand legal scrutiny under the *Daubert* standard.

1. On the vWorkstation desktop, **double-click** the **AccessData FTK Imager icon** to open the FTK Imager application.

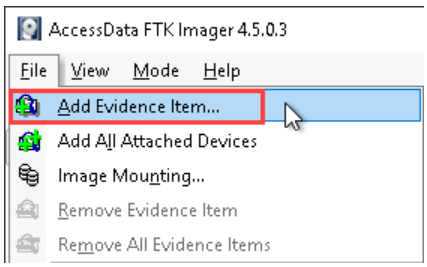


FTK Imager icon

Note: FTK Imager is a free disk imaging and data preview tool developed by AccessData, now owned by Exterro. FTK Imager is a standalone utility associated with the Forensic Toolkit (FTK), a professional-grade digital forensics tool suite that offers enhanced search and analysis functionality for a variety of digital evidence

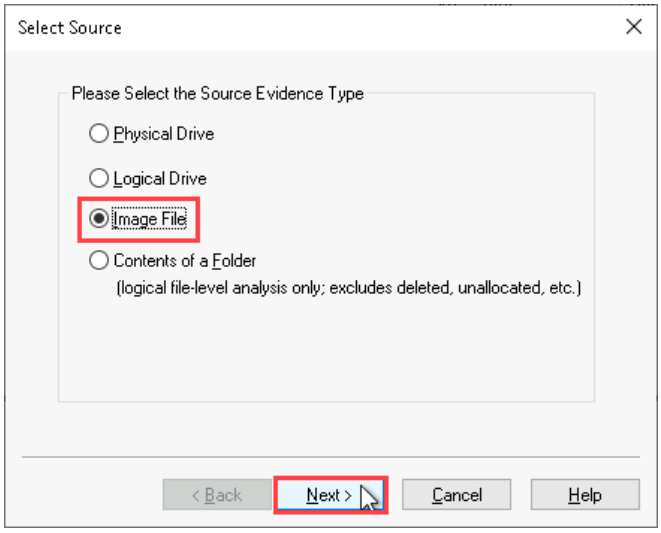
formats. FTK Imager allows you to create forensic images, preview files and folders, mount an image for read-only viewing, recover deleted files, create hashes of files, and generate hash reports. In addition to the normal Graphical User Interface (GUI), certain FTK Imager functions can be run from the command line.

2. In the AccessData FTK Imager menu bar, **select File > Add Evidence Item** to open the Select Source dialog box and begin adding evidence.



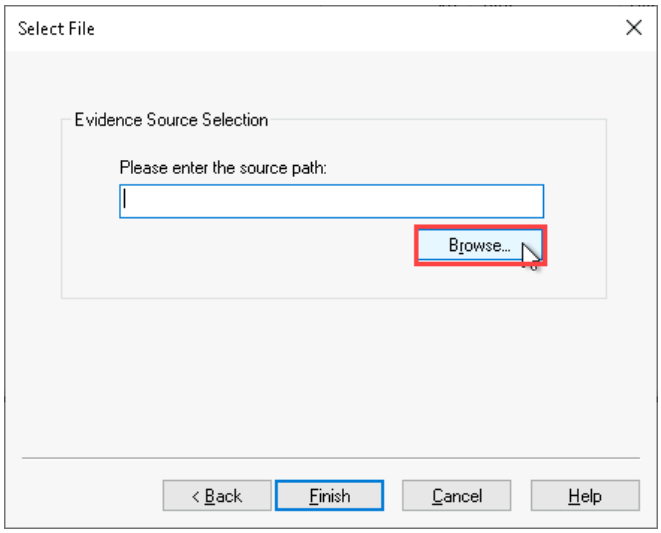
Add Evidence Item menu entry

3. On the Select Source page, **click** the **Image File radio button** and **click Next** to continue.



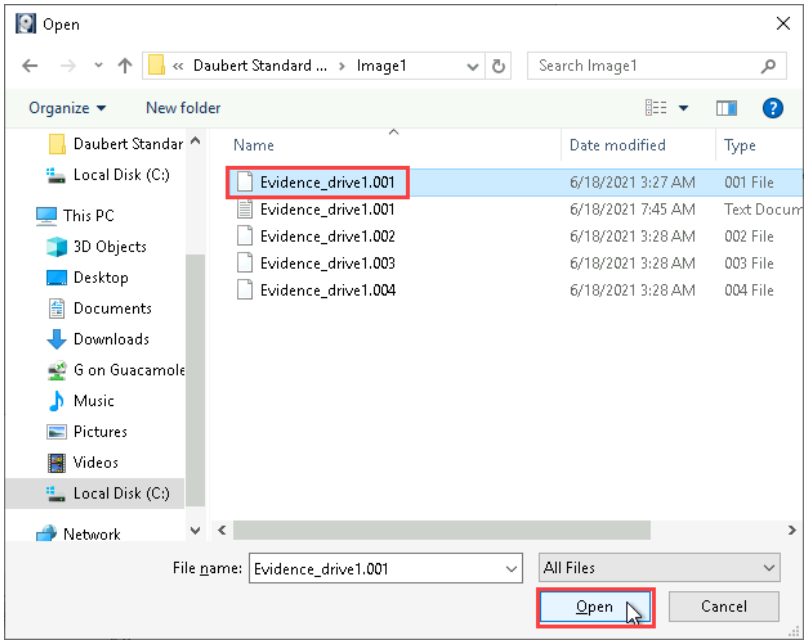
Select Source page

4. On the Select File page, **click** the **Browse button** to open the Open dialog box.



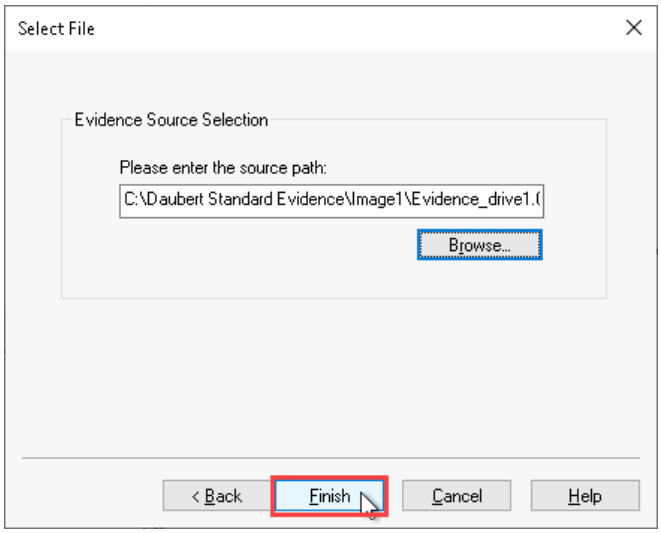
Browse button in the Select File dialog

5. In the Open dialog box, **navigate** to (**This PC > Local Disk (C:) > Daubert Standard Evidence > Image1**), then **select** the **first Evidence_drive1.001 file**, and then **click Open**.



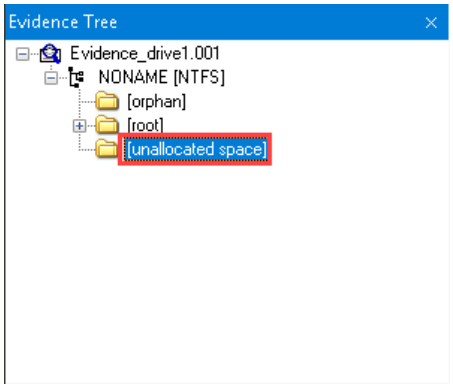
Open the Evidence_drive1.001 file

6. On the Select File page, **click Finish** to open the suspect's drive image in FTK Imager.



Finish adding evidence

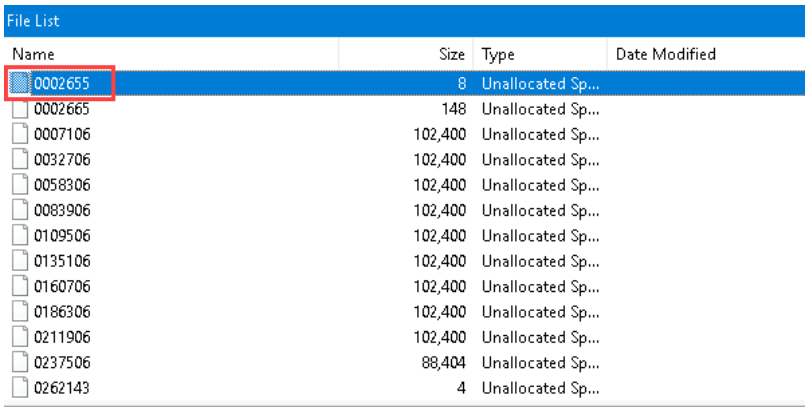
7. In the Evidence Tree pane, **navigate** to **Evidence_drive1.001 item > NONAME [NTFS] > [unallocated space]** to display its contents in the File List pane.



Evidence Tree

Note: In the Windows NTFS file system, deleted files are not actually deleted—they are retained in the unallocated space on the hard drive. This is often a very evidence-rich area of the evidence drive, as deleted files should always prompt investigators to explore motives. For the purposes of your investigation, the unallocated space on Beverly Gates's hard drive seems like a great place to begin looking for evidence.

8. In the File List pane, **select** the **first file (0002655)** to view the contents in the Display pane.



File List

Note: The File List pane displays files in alphabetical order based on the file name. The Display pane will show images, hex view, or clear text, depending on the file type.

9. In the Display pane, **review** the **contents** of the 0002655 file.

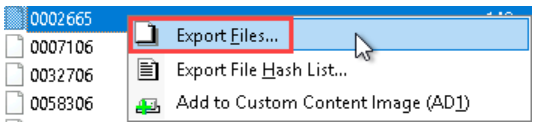
Note: The file appears to be an email between Beverly Gates and several Intricate Solutions employees about setting up a recurring meeting. Nothing suspicious about that—at least outside the context of additional evidence. Time to move to the next file.

10. In the File List pane, **select** the **second file (0002665)** to view the contents in the Display pane.

11. In the Display pane, **review** the **contents** of the 0002665 file.

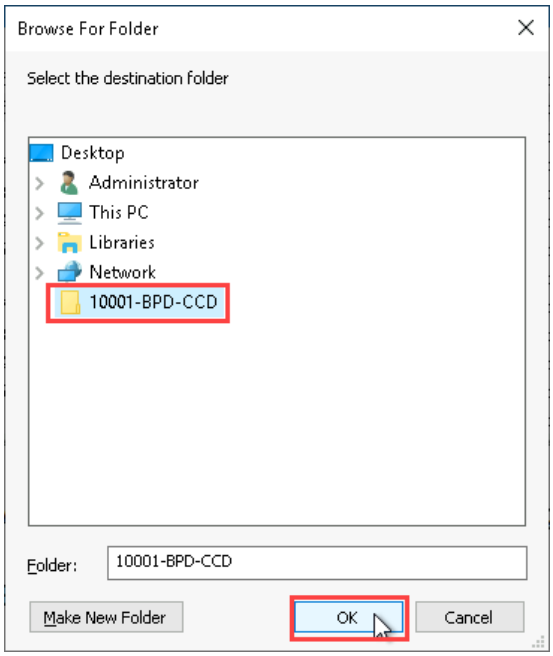
Note: Now this is a little more interesting. Given that Beverly Gates is suspected of being involved in drug trafficking, this file looks an awful lot like an image containing a list of drug dealers, what they sell, and their phone numbers. You would be well-advised to save this file to your evidence folder for future reference.

12. In the File List pane, **right-click** the **0002665 file** and **select Export Files** from the context menu to extract this file as evidence.



Export Files context menu entry

13. In the Save As dialog box, **navigate** to the **10001-BPD-CCD folder** you created earlier in the lab and then **click OK** to save this file to the evidence folder.



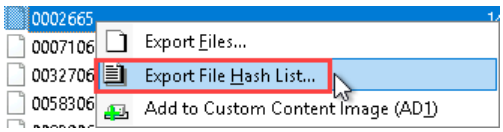
Browse for Folder

14. When prompted, **click OK** to close the Export Results dialog box.



Export Results dialog box

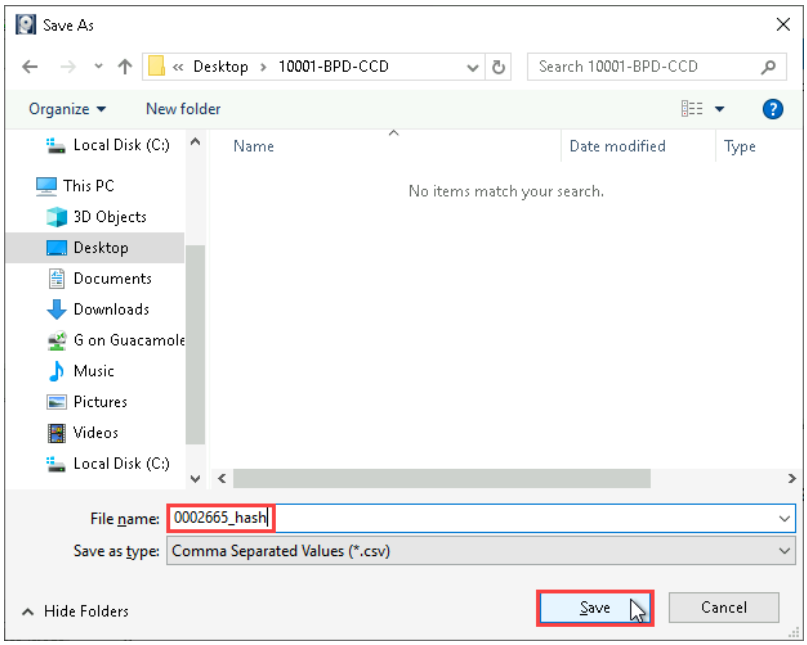
15. In the File List pane, **right-click** the **0002665 file** and **select Export File Hash List** from the context menu to create a hash code for this file.



Export File Hash List context menu entry

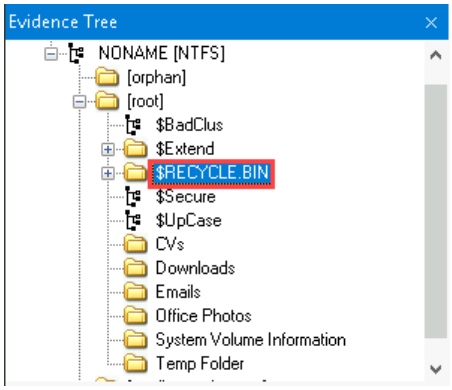
Note: When you use the Export File Hash List function, you are generating a human-readable file that contains three important fields: MD5, SHA1, and filename location. MD5 and SHA1 are hash values for the file and the filename location is the name of the imaged/seized drive along with full-path name (within the drive image) of the file. When vetting and verifying evidence, as long as the file remains unchanged, these three fields will be the same regardless of the forensic tool (FTK Imager, Autopsy, E3, or others) used to view it.

16. In the Save As dialog box, **navigate** to the **10001-BPD-CCD folder (This PC > Desktop > 10001-BPD-CCD)**, then **type** `0002665_hash` in the File name field, and then **click Save** to save this file to the evidence folder.



Save As dialog box

17. In the Evidence Tree pane, **navigate** to **Evidence_drive1.001 item > NONAME [NTFS] > [root] > $RECYCLE.BIN** to view the contents in the Display pane.

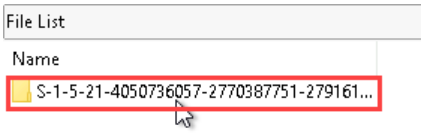


Recycle bin in the Evidence Tree

Note: In Windows systems, the Recycle Bin holds files that the user intends to delete. Like the unallocated space, it is one of the first places an investigator should explore. However, unlike the unallocated space, the contents of the Recycle Bin remain visible to the user and can be easily restored if required. When the Recycle Bin is emptied, any files in the Recycle Bin are "deleted" and moved to the unallocated space.

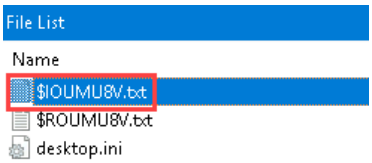
When files are moved to the Recycle Bin in Windows 10, two files are created in the Recycle Bin. The first file is assigned a random alpha-numeric name beginning with $I. This file contains the original location of the deleted file. The second file is assigned the same random name, but beginning with $R. The second file is the actual deleted file. In older versions of Windows (with the exception of Windows Vista), deleted files were renamed DC#.ext, where # is an incrementally increasing integer number (for example, DC1, DC2, DC3, etc.) and ext is the original file's extension (for example, .txt or .exe). The original locations of all deleted files are stored in a single file titled INFO2. For forensic investigators, the fact that a file was deleted suggests that the owner of this computer may have wanted to hide it. This behavior is not inherently criminal, as deleting files from a computer is also simply good hygiene, but in the context of a forensic investigation, it is potentially suspicious.

18. In the File List pane, **double-click** the **S-1-5-21-4050736057-2770387751-2791612479-1001 folder** to open it.



Recycle bin folder in the File List

19. In the File List pane, **select** the **$IOUMU8V.txt file** to view the file contents in the Display pane.



Recycle bin contents in the File List

20. In the Display pane, **review** the contents of the $IOUMU8V.txt file.

Note: The $IOUMU8V.txt file contains the original file name and location of the deleted file, which is now stored as $ROUMU8V.txt in the Recycle Bin. In this case, $ROUMU8V.txt appears to be the Recycle Bin name for a file titled MyRussianMafiaBuddies.txt that was deleted from the Downloads folder.

21. In the File List pane, **select** the **$ROUMU8V.txt file** to view the contents in the Display pane.

Note: As the original file name implies, this file appears to contain a list of Russian names and addresses. Given that Beverly Gates is under investigation for drug trafficking, it would seem reasonable to take the name of this file at face value. You should save this file as another piece of evidence.

22. In the File List pane, **simultaneously press the CONTROL key** and then **click** the **$IOUMU8V.txt file** to select both files, $ROUMU8V.txt and $IOUMU8V.txt.

23. **Repeat steps 12-14** to save the $IOUMU8V.txt and $ROUMU8V.txt files to the 10001-BPD-CCD folder.

24. **Repeat steps 15-16** to export the hash values for the $IOUMU8V.txt and $ROUMU8V.txt files to the 10001-BPD-CCD folder as a .csv file titled **`RecycleBinEvidence_hash`**.

Note: In the next steps, you will attempt to locate copies of the deleted files with their original file names on Beverly Gates's hard drive. FTK Imager does not have a search function, so searching for specific files must be done manually. A thorough investigation using FTK Imager can take days, but for the purpose of this lab, you can look through the folders available in the disk image, as it does not contain a lot of data. Because you know the contents of one deleted file and the original name of the other, you should be able to identify copies of the files fairly easily. At a glance, you should see a few different folders in the root directory, including CVs, Downloads, Emails, Office Photos, System Volume Information, and Temp Folder. As far as folder names go, Temp Folder is a bit strange, which makes it as good a place as any to start.

25. In the Evidence Tree pane, **navigate** to the **001 item > NONAME [NTFS] > [root] > Temp Folder** to display its contents in the File List pane.

Note: Okay, that was a little too easy, but that sure looks like a copy of the MyRussianMafiaBuddies.txt file and an image file titled Nice Guys.png. This seems to be exactly what you're looking for.

26. **Review each file in the Display pane** to verify that they match the files you previously found in the Unallocated space and Recycle Bin.

27. **Repeat steps 12-14** to save the MyRussianMafiaBuddies.txt to the 10001-BPD-CCD folder.

28. **Repeat steps 15-16** to export the hash values for the MyRussianMafiaBuddies.txt file to the 10001-BPD-CCD folder as a .csv file titled **`MyRussianMafiaBuddies_hash`**.

29. **Repeat steps 12-14** to save the Nice Guys.png to the 10001-BPD-CCD folder.

30. **Repeat steps 15-16** to export the hash values for the Nice Guys.png file to the 10001-BPD-CCD folder as a .csv file titled **`Nice guys_hash`**.

31. **Close** the **FTK Imager window**.

32. On the vWorkstation Desktop, **double-click** the **10001-BPD-CCD folder** to open it.

33. In the 10001-BPD-CCD folder, **double-click** the **0002665_hash.csv file** to open the file in the Notepad application.

34. **Maximize** the **Notepad window** to ensure that the MD5, SHA1, and file name are fully visible.

    If necessary, **select Notepad > Format > Enable Word Wrap.**

Note: The MD5 and SHA1 hash codes generated by FTK Imager will remain the same, no matter which investigator, which program, or which day the files are touched. The hash codes will change only if the file itself is changed, thereby assuring the court that the evidence they are looking at has not been altered.

35. **Make a screen capture** showing the **contents of the 0002665_hash.csv file**.

36. **Close** the **Notepad window.**

37. **Repeat steps 32-35** for each .csv in the case folder.

38. **Make a screen capture** showing the **contents of the RecycleBinEvidence_hash.csv file**.

39. **Make a screen capture** showing the **contents of the MyRussianMafiaBuddies_hash.csv file**.

40. **Make a screen capture** showing the **contents of the Nice guys_hash.csv file**.

41. **Close** the **10001-BPD-CCD folder**.

Part 3: Verify Hash Codes with E3

In this part of the lab, you will use Paraben's Electronic Evidence Examiner (E3) to verify the MD5 hashes generated by FTK. E3 is a forensic investigation tool designed to support the entire process of digital evidence handling, from acquisition to analysis. While it offers a wide range of features—such as evidence triage, email analysis, and smartphone forensics—the focus here will be on hash code verification, which is a critical step in ensuring evidence integrity.

By verifying the hash codes with E3, you are confirming that the files you are working with have not been altered or tampered with, which is essential for maintaining the integrity of the evidence throughout the investigation and meets the intent of Learning Objective 5.

In real-world investigations, the process of evidence collection and analysis is often divided between multiple individuals. For instance, a first responder may acquire the evidence using tools like FTK Imager, while a forensic analyst will later process the evidence using different software. Since evidence can sit for lengthy periods of time before being analyzed, it's vital to verify the hash codes to ensure that no tampering occurred during transit or while in storage. As mentioned previously, failing to verify hash codes could result in the dismissal of evidence in court, potentially compromising the entire case.

In the next steps of this lab, you will acquire the same evidence from Part 2, generate new hash codes, and compare them to the hash codes you generated earlier with FTK. This step-by-step process reinforces the importance of hash verification in maintaining the authenticity of digital evidence and ensuring that it will hold up under legal scrutiny.

For a deeper dive into E3's full capabilities, feel free to explore its features in the E3 Brochure, but remember that the main focus in this task is the critical process of hash code verification. not been altered in transit.

In the next steps, you will acquire the same evidence used in Part 2, generate hash codes, and compare them to the hash codes generated by FTK.

1. On the vWorkstation desktop, **double-click** the **Electronic Evidence Examiner icon** to open the E3 application.



E3 icon

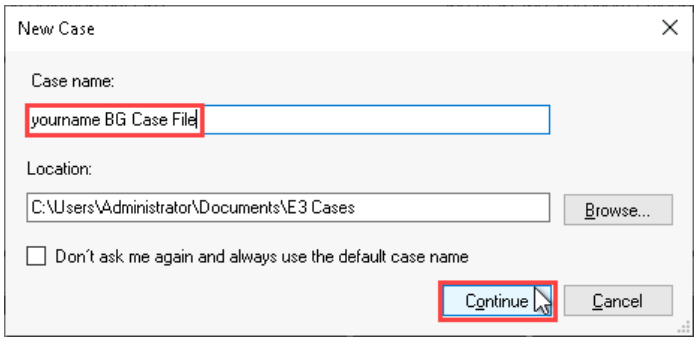
Note: E3 may take several minutes to load. The E3 welcome page opens with shortcuts to the most common activities that forensic investigators will perform within the tool.

2. On the Welcome page, **click** the **Add Evidence button** to open the New Case dialog box.



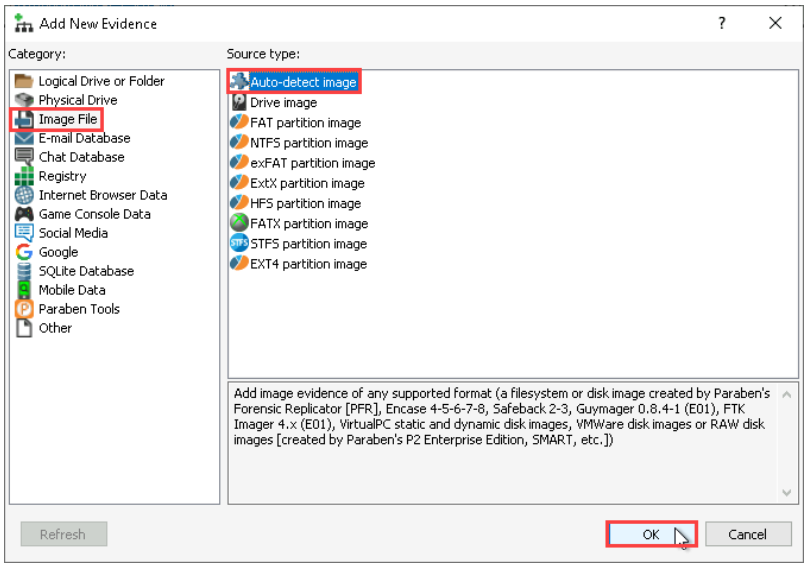
Welcome page - Add Evidence

3. In the New Case dialog box, **type** ***yourname*** **`BG Case File`** in the Case name field, replacing *yourname* with your own name, and then click Continue to create your new case file and open the Add New Evidence dialog box.



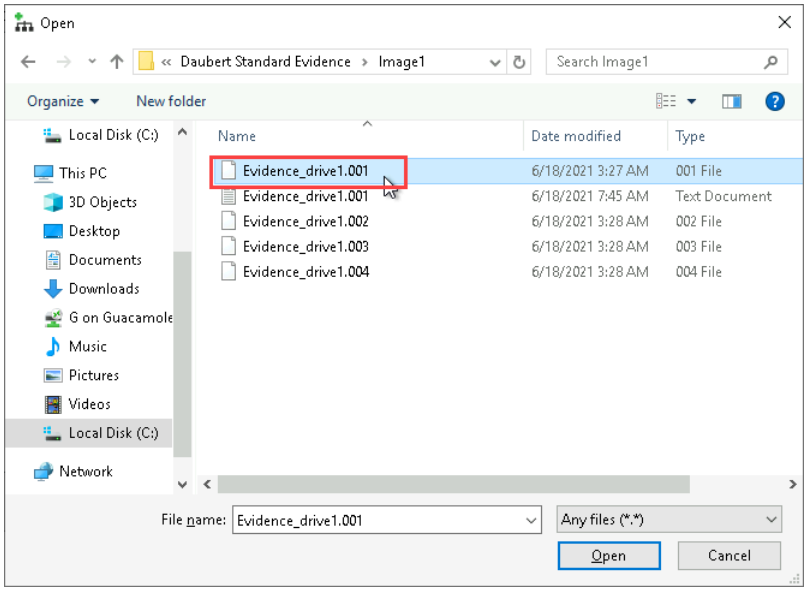
New Case dialog box

4. In the Add New Evidence dialog box, **click** the **Image File category**, then **select** the **Auto-detect image Source type** and **click OK** to continue.



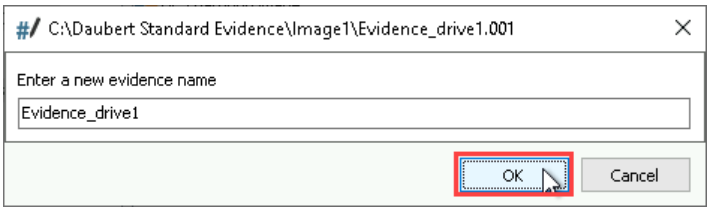
Add New Evidence - Auto-detect image

5. In the Open dialog box, **navigate** to **This PC > Local Disk (C:) > Daubert Standard Evidence > Image1** and **double-click** the first **Evidence_drive1.001 file** to select the digital drive image for this lab.



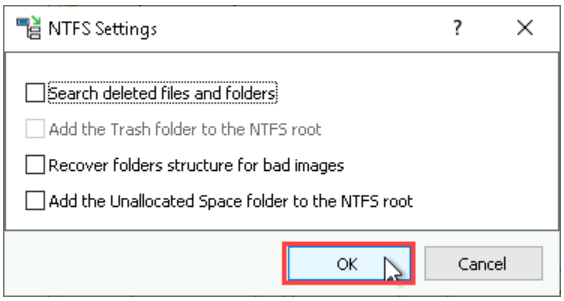
Open dialog box

6. When prompted, **click OK** to accept the default name for the drive image and add the data from the drive image to your case file.



Evidence name

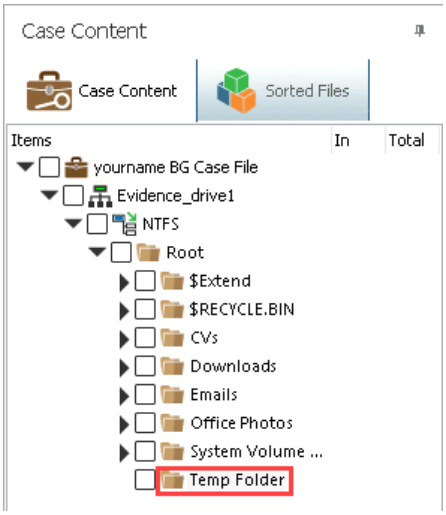
7. When prompted, **click OK** to close the NTFS Settings dialog box.



NTFS Settings dialog box

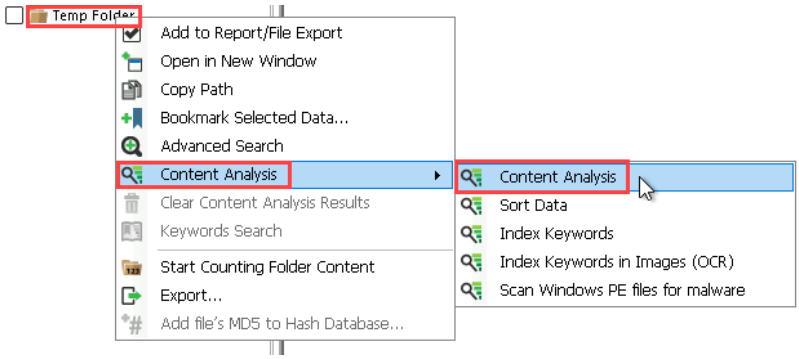
Note: The yourname BG Case File case will appear in the Case Content pane. The Case Content pane is the primary display and navigation area for all evidence in the open case file. Within the Case Content pane's tree-view structure, you can access case nodes, evidence nodes, evidence type nodes, folder nodes, and data grids/streams. Technically, the Case Content pane does not store the actual evidence. Instead, it provides a series of links to elements within the physical evidence. When you select a specific node or grid within the Case Content pane, the contents of the enclosing folder or database will be displayed in the Data Viewer in the center pane. Within the Data Viewer, you can select individual files, folders, and records. Additional information about the currently selected node, grid, file, folder, or record will be displayed in the Viewers pane on the right, which provides multiple ways to view your current selection.

8. In the Case Content pane, **navigate** to ***yourname* BG Case File > Evidence_drive1 > NTFS > Root > Temp Folder** to display the contents in the center pane.



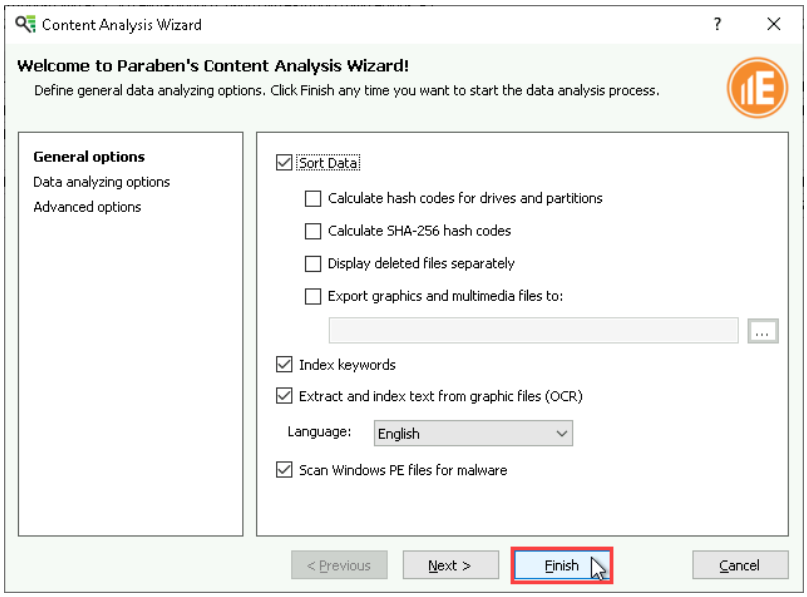
Temp Folder in the Case Content pane

9. In the Case Content pane, **right-click** the **Temp Folder node** and **select Content Analysis > Content Analysis** from the context menu to open the Content Analysis wizard.



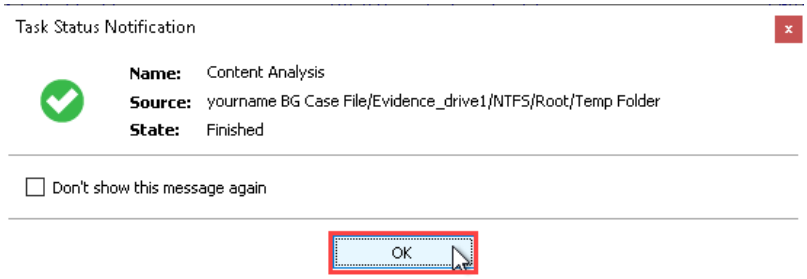
Content Analysis context menu entry

10. In the Content Analysis wizard, **click Finish** to run the content analyzer with default settings.



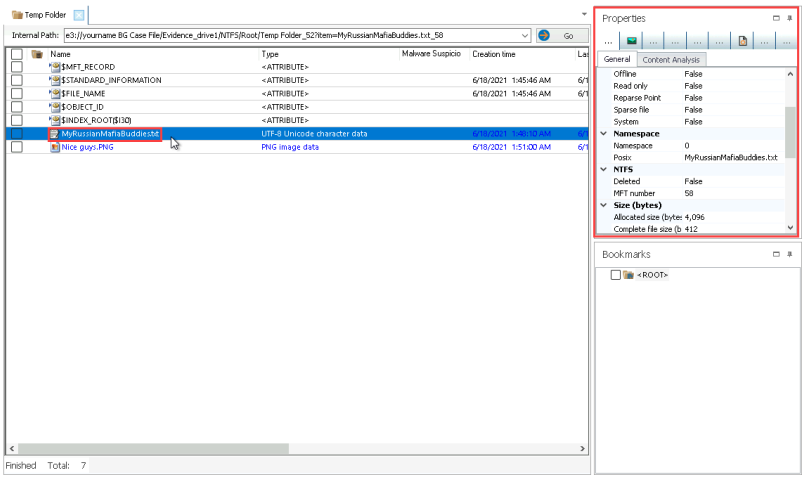
Content Analysis wizard

11. When prompted, **click OK** to close the Task Status Notification dialog box.



Task Status Notification dialog box

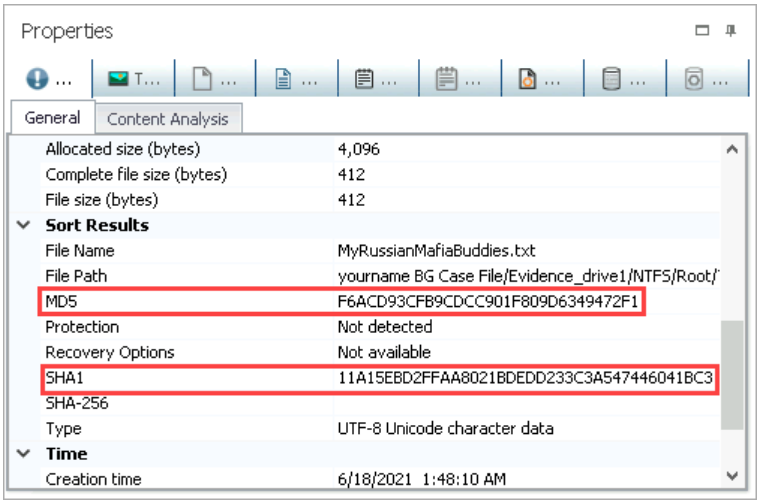
12. In the center pane, **select** the **MyRussianMafiaBuddies.txt file** to display information about the file in the Properties pane.



MyRussianMafiaBuddies.txt properties

13. In the Properties pane, **scroll down** to reveal the MD5 and SHA1 values.

    If necessary, **expand** the **Properties pane.**



MD5 and SHA1 values

14. **Make a screen capture** showing the **MD5 and SHA1 values for the MyRussianMafiaBuddies.txt file**.

15. **Repeat steps 11-12** for the Nice Guys.png file.

16. **Make a screen capture** showing the **MD5 and SHA1 values for the Nice Guys.png file**.

17. **Describe** how the hash values produced by E3 for the incriminating files compare to those produced by FTK. Do they match?

18. **Close** the **E3 window**.

Note: This concludes Section 1 of the lab.

Applied Learning

Note: SECTION 2 of this lab allows you to apply what you learned in SECTION 1 with less guidance and different deliverables, as well as some expanded tasks and alternative methods. You will explore the suspect's drive image to locate additional evidence. You will also use Autopsy, another popular digital forensics tool, to validate the hash codes created during your preliminary investigation.

Part 1: Extract Evidence Files and Create Hash Codes with FTK Imager

In this part of the lab, you will continue your investigation into the seized drive image of Beverly Gates, applying the skills you've developed so far while also taking more initiative. This section builds on the tasks you completed in Section 1, and now you are expected to work more independently as you delve deeper into the evidence. Specifically, you will search for email evidence that could potentially link Beverly to criminal activity or collaborators in the drug trafficking operation.

Your supervisor has reason to believe that Beverly Gates may have exchanged incriminating emails with her partners, so your task is to locate these emails within the drive image. As part of your investigation, you will also demonstrate the efficacy of hash codes by intentionally altering a sample evidence file. This will allow you to see firsthand how hash codes change when the file is modified. You will then re-add the altered file into FTK Imager, generate and export new hash codes, and understand how this step is directly tied to the Daubert standard. The ability to prove that evidence has remained unaltered from seizure to court presentation is essential in ensuring its reliability and admissibility in court. In addition to finding email evidence, this task is directly tied to, and satisfies the intent of, Learning Objectives 4 and 5.

Think critically about the task. How could email evidence be hidden or disguised on a hard drive? What might the implications be if you find that evidence has been altered? These are questions you'll need to consider as you proceed, as altered or hidden evidence could seriously undermine the integrity of your investigation and any legal proceedings.

As you re-open the drive image in FTK Imager, your specific goal is to carefully search for email files that might link Beverly Gates to her criminal activities or collaborators. This is a crucial step in the Beverly Gates investigation, where finding relevant communications could play a significant role in proving or disproving the allegations against her. By applying the skills you've learned so far, you'll be contributing to the investigation while ensuring that all evidence is properly handled, validated, and prepared for potential use in court.

1. From the vWorkstation desktop, **launch** the **AccessData FTK Imager application**.

2. **Add** the **Evidence_drive1.001 drive image** as evidence.

3. In the Evidence Tree, **navigate** to **Evidence_drive1.001 > NONAME [NTFS] > root > Emails**.

4. **Identify** a **suspicious email file** in the Emails folder.

   Hint: It's the one that references stardust.

5. **Make a screen capture** showing the **contents of the suspicious email file in the Display pane**.

6. **Export** the **suspicious email file** to the 10001-BPD-CCD folder.

7. **Export** the **File Hash List** for the suspicious email file to the 10001-BPD-CCD folder.

8. **Minimize** the **FTK Imager window**.

9. From the 10001-BPD-CCD folder, **open** the **suspicious email file** in Notepad.

10. In Notepad, **delete** the **body of the suspicious email**, then **close** the file and **save** your changes.

11. **Restore** the **FTK Imager window**.

12. **Add** the **edited suspicious email file** as evidence.

    Hint: You can add the entire 10001-BPD-CCD folder as evidence.

13. **Export the File Hash List** for the edited suspicious email file to the 10001-BPD-CCD folder.

14. **Close** the **FTK Imager window**.

15. From the 10001-BPD-CCD folder, **open** the **csv files** for both the original and edited suspicious email file in Note pad, then **arrange** the **windows** side by side.

16. **Make a screen capture** showing the **two hash values for the suspicious email file**.

Note: The hash values for the two files should not match, demonstrating that changes made to an evidence file will result in a different hash code.

17. **Close** any **open windows**.

Part 2: Verify Hash Codes with Autopsy

In this part of the lab, you will apply what you learned about hash validation in a more independent context by using Autopsy, a popular tool in digital forensics. This builds on the skills you developed in Section 1 with FTK Imager and reinforces your ability to use different tools for evidence validation, improving your understanding of their practical application. Autopsy is a graphical front-end for The Sleuth Kit, an open-source collection of command-line utilities used to extract and analyze data from drive images. Unlike FTK Imager, which you used in the previous section, Autopsy offers a different interface and toolset, allowing you to explore how different tools can complement each other to strengthen the reliability of the evidence validation process.

In this step, you will use Autopsy to validate the hash code for the suspicious email file you extracted from the suspect's drive image in Part 1. Just like you did earlier, you will generate and validate hash codes, but now you're applying this process in a new context with a different tool. By doing this, you will not only reinforce your previous work but also extend your skills, preparing you for real-world scenarios in which multiple tools are used to validate evidence.

Tools like Autopsy are commonly used in actual digital forensic investigations, especially when validating evidence integrity. The ability to integrate community-developed or custom modules with Autopsy for specific cases further enhances the tool's capability, just as it will help you during this lab. Validating the suspicious email file in the Gates investigation could be crucial in proving Beverly Gates’s involvement in illegal activities. Ensuring the file’s integrity means it can be used in court without question, supporting the case against her.

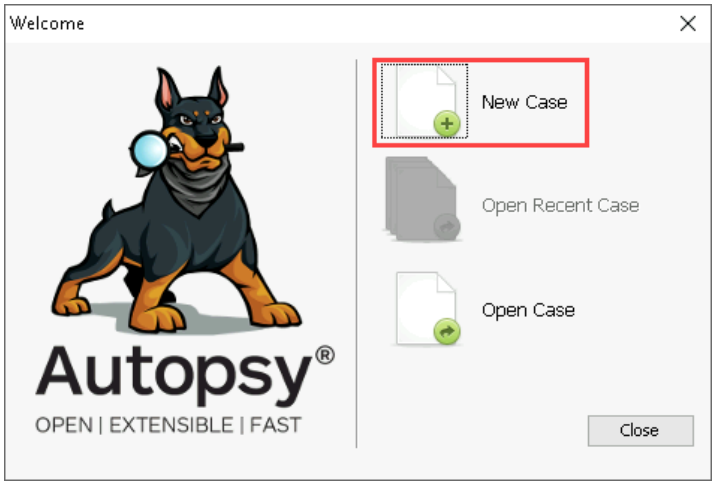
As you move forward, consider why you're switching from FTK Imager to Autopsy for this task. By diversifying the tools you use, you’re not only strengthening your understanding of the validation process but also gaining experience with the variety of tools that real digital forensic analysts rely on to ensure that evidence remains intact and reliable throughout the investigation.

1. From the vWorkstation desktop, **launch** the **Autopsy application**.



Autopsy icon

2. In the Autopsy Welcome window, **click** the **New Case option** to create a new case file.



Welcome window

3. On the Case Information page, **enter** `BG_evidence` as the case name and **select** the **Documents folder** as the Base Directory, then **click Next** to continue.

4. On the Optional Information page, **click Finish** to save your case file and open the Add Data Source Wizard.

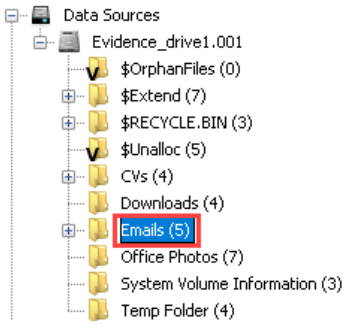
5. On the Select Type of Data Source To Add page, **click Next** to accept the default (Disk Image or VM File) and continue.

6. On the Select Data Source page, **click** the **Browse button**, then **navigate** to **This PC > Local Disk (C:) > Daubert Standard Evidence > Image1** and **double-click** the **Evidence_drive1.001 file**, and then **click Next** to continue.

7. On the Configure Ingest Modules page, **click Next** to accept the default settings and continue.

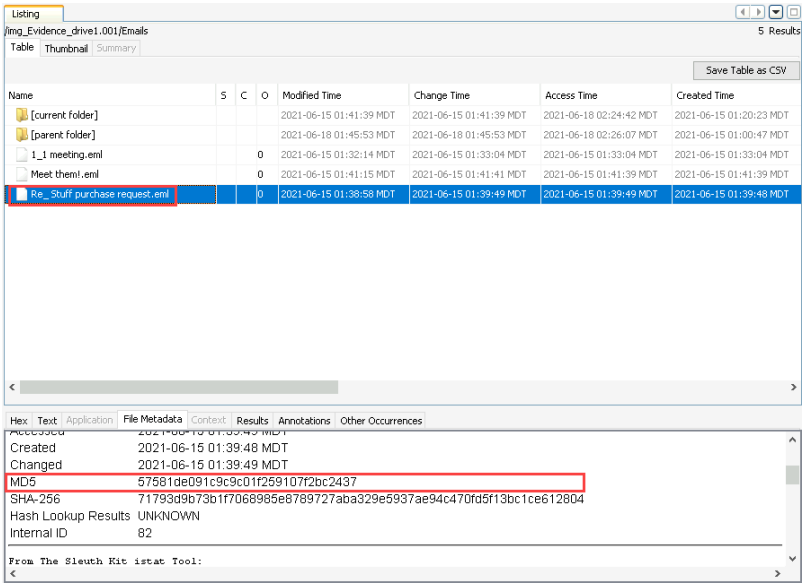
8. On the Add Data Source page, **click Finish** to continue.

9. In the Tree Viewer, **navigate** to **Data Sources > Evidence_drive1.001 > Emails** to display the contents in the Result Viewer.



Emails folder in the Tree Viewer

10. In the Result Viewer, **select** the **Re_Stuff purchase request.eml file**, then **click** the **File Metadata tab** and **scroll down** to display the MD5 hash value.



Result Viewer

11. **Make a screen capture** showing the **MD5 field in the Result Viewer**.

12. **Describe** how the hash value produced by Autopsy compares to the values produced by FTK Imager for the two .eml files.

13. **Close** the **Autopsy window**.

Part 3: Verify Hash Codes with E3

In this part of the lab, you will apply your knowledge of multiple forensic tools independently to validate evidence, a process that mirrors real-world digital forensics practices. This step builds on what you've learned in Section 2 and reflects the multi-tool approach commonly used in the field to ensure the reliability and accuracy of evidence. By using Paraben's E3 in addition to FTK Imager and Autopsy, you are performing cross-validation, which strengthens the reliability of the evidence you're working with. In forensic investigations, it's common to validate evidence with multiple tools to ensure that no single tool's limitations or potential errors affect the findings.

In this case, you'll focus on validating the hash code of the suspicious email file found in the Beverly Gates case. This task is critical because validating the integrity of this file ensures that it has not been altered, which is vital for its admissibility in court. Without proper hash validation, any alteration could lead to questions about the authenticity of the evidence, potentially undermining the investigation. Hash validation not only protects the evidence's integrity but also aligns with Learning Objective 5, which emphasizes understanding how to maintain evidence integrity using validated forensic tools and techniques. By using multiple tools, you gain a deeper understanding of how different tools can work together to provide reliable results, reinforcing the principles of the *Daubert* standard.

Take a moment to reflect on why it's essential to validate the same file with multiple tools. Could discrepancies between the tools indicate potential issues with one of them? How does cross-validation enhance your confidence in the evidence? This exercise will encourage you to critically think about the tools and techniques you use, reinforcing the importance of reliability and consistency in forensic investigations.

1. From the vWorkstation desktop, **launch** the **E3 application**.

2. From the Welcome screen, **open** the ***yourname* BG Case File** that you created in Section 1.

3. From the Case Content tree, **navigate** to the **Emails folder** and **locate** the **suspicious email file** that you identified in FTK Imager and Autopsy.

4. **Right-click** the **suspicious email file** and use the **Export function** to save the file to the 10001-BPD-CCD folder.

5. **Close** the **E3 window**.

6. From the 10001-BPD-CCD folder, **open** the **MD5 file** for the suspicious email file in Notepad.

7. **Make a screen capture** showing the **MD5 value produced by E3**.

8. **Describe** how the hash value produced by E3 compares to the values produced by FTK Imager for the two .eml files and the value produced by Autopsy.

9. **Close** the **Notepad window**.

Note: This concludes Section 2 of the lab.

Challenge and Analysis

Note: The following exercises are provided to allow independent, unguided work using the skills you learned earlier in this lab—similar to what you would encounter in a real-world situation.

Part 1: Verify Hash Codes on the Command Line

In this lab, you're continuing to take on the role of a forensic specialist investigating evidence associated with a drug trafficking case involving Beverly Gates. While using various forensic tools, you've encountered the FTK Imager and learned that its functions can also be run from the command line. This is an important skill to develop, as command-line tools are often favored by forensic specialists for their automation, efficiency, and flexibility. Command-line tools can be particularly useful in resource-limited environments, where a graphical interface might not be available or practical.

As you move forward, you'll need to take the initiative by researching how to use FTK Imager's command-line version. In this part of the lab, you will research the command used to generate SHA1 and MD5 hashes for a specific file. This task is designed to encourage independent research, helping you grow in your ability to navigate and apply forensic tools.

To complete this task, you'll need to:

- Research the FTK Imager command-line documentation online. Look for reliable sources like official product manuals, trusted forums, or technical blogs that cover FTK Imager CMD commands.
- Identify the specific command used to generate SHA1 and MD5 hashes for a file.
- Open a Command Prompt window, navigate to the FTK Imager CMD tool location on your machine (C:\Program Files\AccessData\FTK Imager\cmd), and use the command you identified to verify the SHA1 and MD5 hashes of the Evidence_drive1.001 file used in this lab.

As you research and apply this knowledge, you might find yourself reflecting on why some digital forensic analysts prefer command-line tools over graphical interfaces (or vice versa). Document your thoughts on this in

your lab report, considering factors like speed, precision, and control in forensic investigations.

Although this section focuses on building your skill set, remember that in real-world investigations, forensic specialists are often required to use command-line tools in various environments, especially when working with large data sets or on resource-limited systems. Becoming proficient with these tools will not only enhance your current forensic abilities but also improve your career prospects in the digital forensics field.

By building on your knowledge of command-line tools, you're expanding your forensic toolkit and preparing for the increasing independence required in your later investigative tasks. These skills are critical for professional development, making you a more versatile and capable forensic investigator.

Make a screen capture showing the **hash values for the Evidence_drive1.001 file**.

Part 2: Locate Additional Evidence

New information has emerged during the investigation, suggesting that there may be additional evidence located on the seized drive image. This new lead likely stems from an anomaly discovered in previous hash analysis or may have surfaced through a new tip or investigative clue, helping to guide the current phase of the investigation. In real-world digital forensics, investigators often revisit previously collected evidence when new leads arise, demonstrating the iterative nature of the process.

Using any of the three forensic programs provided in this lab (FTK Imager, Autopsy, or E3), your task is to identify the initial location and names of the following files in the Evidence_drive2 disk image (located within the Daubert Standard Evidence / Image2 folder):

- $R354ELH.xlsx
- $RBQEOTL.doc
- $RX3177E.pdf

Define the original file names and file paths for each file.

As you proceed, consider why you chose the forensic tool you did, whether it's based on ease of use, specific features, or familiarity. This encourages critical thinking in your selection process, as each forensic tool offers unique strengths for different stages of analysis.

Note: This concludes Section 3 of the lab.